

EXPERIMENT 10

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Question 1

Question:

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction. If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using **SAVEPOINTS**. The system should display clear messages for successful and failed insertions.

Solution:

DO \$\$

BEGIN

BEGIN

-- Insert multiple students

INSERT INTO students(name, age, class)

VALUES ('Anisha', 16, 8);

INSERT INTO students(name, age, class)

VALUES ('Neha', 17, 8);

INSERT INTO students(name, age, class)

VALUES ('Mayank', 19, 9);

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-- If all succeed

RAISE NOTICE 'Transaction Successfully Done';

EXCEPTION WHEN OTHERS THEN

-- If any insert fails

RAISE NOTICE 'Transaction Failed... Rolling back changes.';

RAISE;

END;

END;

\$\$;

To check the inserted records:

SELECT * FROM students;

Using SAVEPOINT

For preserving previous successful inserts while rolling back only the failed insertion:

BEGIN;

INSERT INTO students(name, age, class)

VALUES ('Anisha', 16, 8);

SAVEPOINT sp1;

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```
INSERT INTO students(name, age, class)
VALUES ('Neha', 17, 8);
```

```
SAVEPOINT sp2;
```

```
-- If the next insertion fails
```

```
INSERT INTO students(name, age, class)
VALUES ('Anisha', 17, 8);
```

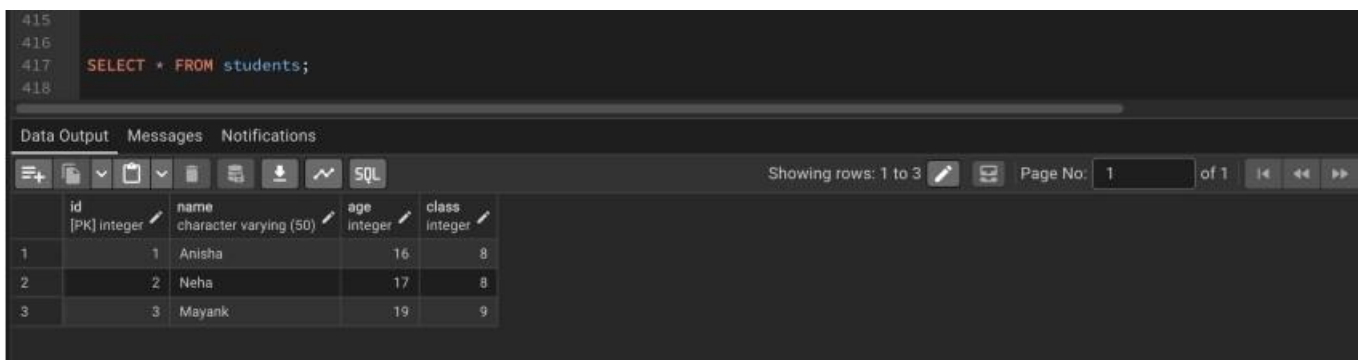
```
-- Roll back only the failed part
```

```
ROLLBACK TO SAVEPOINT sp2;
```

```
COMMIT;
```

Here, the previously successful records remain in the transaction, while the failed insertion is rolled back to the latest savepoint.

The PDF also demonstrates the **rollback case** where inserting a duplicate name violates the students_name_key constraint and produces the message “**Transaction Failed... Rolling back changes.**”



415
416
417 `SELECT * FROM students;`
418

Data Output Messages Notifications

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	id [PK] integer	name character varying (50)	age integer	class integer
1	1	Anisha	16	8
2	2	Neha	17	8
3	3	Mayank	19	9

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```
420
421 --VIOLATED SCENARIO
422 DO $$
423 BEGIN
424     -- Start a transaction
425     BEGIN
426         -- Insert multiple students
427         INSERT INTO students(name, age, class) VALUES ('Anisha',16,8);
428         INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);
429         INSERT INTO students(name, age, class) VALUES ('Anisha',17,8); --ERROR
430         INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);
431
432         -- If all succeed
433         RAISE NOTICE ' Transaction Successfully Done!';
434
435     EXCEPTION WHEN OTHERS THEN
436         -- If any insert fails
437         RAISE NOTICE 'Transaction Failed..! Rolling back changes.';
438         RAISE; -- this will rollback the entire transaction
439     END;
440 END;
441 $$;
442
```

Data Output Messages Notifications

NOTICE: Transaction Failed..! Rolling back changes.

ERROR: duplicate key value violates unique constraint "students_name_key"
Key (name)=(Anisha) already exists.

SQL state: 23505

Detail: Key (name)=(Anisha) already exists.

Context: SQL statement "INSERT INTO students(name, age, class) VALUES ('Anisha',16,8)"
PL/pgSQL function inline_code_block line 6 at SQL statement